1. Problem Statement: Many students struggle to prepare for standardized exams like the SAT, ACT, and AP exams because high-quality tutoring and realistic practice can be expensive or limited. I want to create a free platform that provides realistic practice questions, identifies students' weak areas, and helps them understand their mistakes through personalized explanations and targeted practice. My goal is to make high-quality test preparation more attainable for all students while reducing the stress and burnout that often come with exam prep.

2. Project Statement: I plan to build a free Java-based SAT preparation program that provides realistic practice questions for both Math and Reading & Writing, identifies students' weaknesses, and helps them improve through personalized explanations and targeted practice. Similar to Elite Prep, but with a more accessible and personalized approach, my goal is to provide high-quality test preparation completely free while tracking students' progress and eventually expanding to ACT and AP exams.

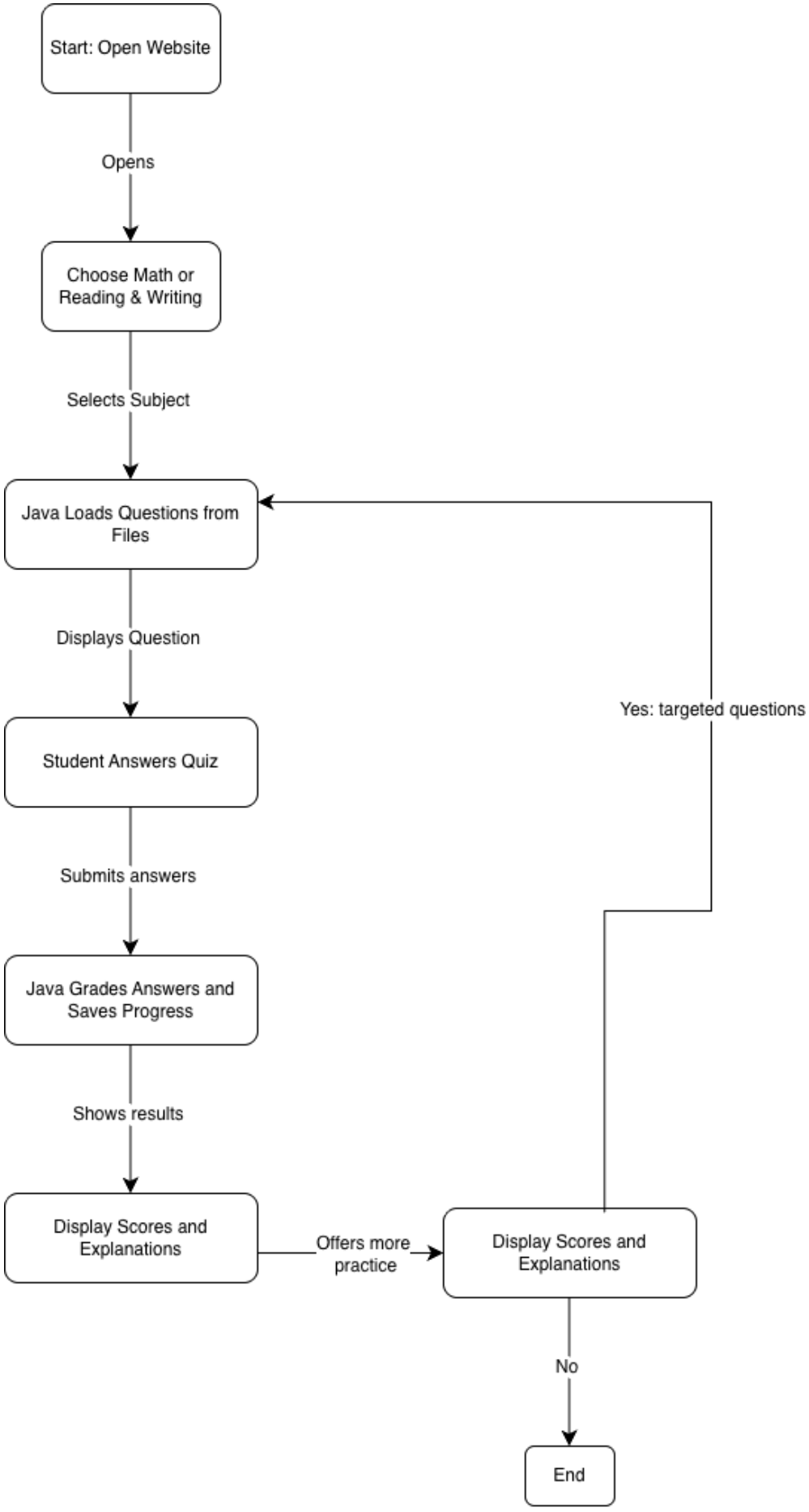
3. Tech Stack:
    a. Programming Language:
        i. Java for the backend, including quiz management, grading, personalized practice, and student progress tracking.
        ii. HTML to structure the website and display practice questions, answer choices, and results.
        iii. CSS to design and style the website's interface.
        iv. Bash to compile and run the Java program and start the server.
    b. Libraries and Tools:
        i. Java's ArrayList and arrays to store and organize practice questions, answers, and student results.
        ii. Java's built-in file I/O (BufferedReader/BufferedWriter & PrintWriter) to read questions and save student progress.
        iii. Java's Scanner class for reading input during console-based testing.
        iv. Object-Oriented programming to organize the questions, quizzes, and student information.
        v. Git and GitHub for version control and tracking project development.
    c. Data Storage:
        i. Text or CSV files to store original SAT Math and Reading & Writing questions, correct answers, explanations, and difficulty levels.
        ii. Separate files to save students' quiz scores, incorrect answers, and progress across practice sessions.
        iii. The MVP will not require an external database.
    d. External Data Sources:

    i. Publicly available College Board resources, including Bluebook practice tests and the SAT Question Bank, will be used as references for the exam format, topics, and difficulty levels.
    ii. Practice questions and explanations will be original rather than copied from official exams.
    iii. No external APIs will be required for the MVP, since the question bank will be stored locally.
4. Milestones:
    a. 1. Set up the GitHib repository and organize the project files.
    b. 2. Build a Java server using ServerSocket and Socket to display HTML pages.
    c. 3. Design the website interface using HTML and CSS.
    d. 4. Create a question bank for SAT Math and Reading & Writing.
    e. 5. Implement the quiz system and automatic grading.
    f. 6. Add explanations for incorrect answers.
    g. 7. Develop targeted practice based on students' weak areas.
    h. 8. Save student scores and progress using Java file I/O.
    i. 9 . Test the website, fix bugs, and finalize the README.

5. Matrix:

	Urgent	Not Urgent
Important	DO - MVP - Create an interactive website with SAT-style practice quizzes for both Math and Reading & Writing using a bank of original questions. - Automatically grade answers and identify the topics or skills students struggle with. - Provide explanations for incorrect answers and additional practice questions targeting those weaknesses. - Track students' scores and progress across practice sessions.	SCHEDULE - (For V2.0+) - Introduce AI-generated practice questions to provide a larger variety of realistic exam-style problems, with answer verification. - Develop full-length timed practice exams with adaptive difficulty similar to the digital SAT format. - Expand the platform to support ACT and AP exam prep. - Add a personalized study plan and interactive AI tutoring.
Not Important	DELEGATE - (Use a library) - Use Java's Scanner class to handle user input. - Use Java's ArrayList and arrays to store and manage practice questions and results. - Use Java's built-in file reading and writing classes to load questions and save student progress instead of creating a custom storage system. - Use HTML and CSS.	DELETE - Cut it - Live video tutoring session with human tutors. - Webcam monitoring and proctoring during practice exams. - College application and essay-writing assistance unrelated to exam prep.

6. Link to repo: https://github.com/hariniinidumoluu/Sat-Prep-Platform


Start: Open Website
Opens
Choose Math or Reading & Writing
Selects Subject
Java Loads Questions from Files
Displays Question
Student Answers Quiz
Submits answers
Java Grades Answers and Saves Progress
Shows results
Display Scores and Explanations
Offers more practice
Display Scores and Explanations
Yes: targeted questions
No
End

7.